

SAT Hard Question Set 9

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Novel Prep

Overview

Question 1

Elizabeth has \$4,500 in an account. Each year she expects to have **11% more** money in the account than the previous year.

Which of the following models best describes how Elizabeth expects the money in her account to change over time?

- A. Decreasing exponential
- B. Decreasing linear
- C. Increasing exponential
- D. Increasing linear

Answer 1

Correct Answer: C

Question 2

The function

$$T(x) = \frac{20,000 - 6.5x}{1,000}$$

gives the estimated air temperature $T(x)$, in degrees Celsius ($^{\circ}\text{C}$), at an altitude of x meters.

If the estimated temperature surrounding the hot air balloon is 15°C , which of the following is closest to the altitude x of the hot air balloon?

- A. 20
- B. 769
- C. 1,333
- D. 5,385

Answer 2

Correct Answer: B

Question 3

The function f is defined as:

$$f(x) = (x - a)(x - b)$$

where a and b are integer constants.

Given that:

$$f(17) > 0, \quad f(20) < 0, \quad f(23) > 0$$

What is one possible value of $a + b$?

- A. 39
- B. 38
- C. 37
- D. 36

Answer 3

Correct Answer: A

Question 4

The positive number a is 2,106% of the sum of the positive numbers b and c , and b is 81% of c .

What percent of b is a ?

- A. 4,706%
- B. 47.06%
- C. 470,600%
- D. 47,060%

Answer 4

Correct Answer: A

Question 5

For the function f , for every increase of 2 in the value of x , the value of $f(x)$ increases by a factor of c , where c is a constant.

Which of the following equivalent forms of function f displays the value of c as the base or the coefficient?

A. $f(x) = 42(2)^{6x}$

B. $f(x) = 42(8)^{2x}$

C. $f(x) = 42(64)^x$

D. $f(x) = 42(4096)^{x/2}$

Answer 5

Correct Answer: D

Question 6

A study investigated the effect of magnesium supplementation on sleep quality. 190 adults were randomly selected, and their sleep efficiency was measured. Participants were randomly assigned to take a magnesium supplement or a placebo for 6 weeks. The study concluded that magnesium improved sleep quality. What feature of this study supports this conclusion?

- A. There were more than 100 participants in the study.
- B. The participants were selected at random from the community center.
- C. The sleep efficiency of each participant was measured again at the end of 6 weeks.
- D. Each participant was randomly assigned to take a magnesium supplement or a placebo.

Answer 6

Correct Answer: D

Question 7

A circle in the xy -plane has its center at $(-1, 1)$. Line t is tangent to this circle at the point $(7, -6)$.

Which of the following points also lies on line t ?

- A. $(0, 7)$
- B. $(6, 9)$
- C. $(14, 2)$
- D. $(15, 1)$

Answer 7

Correct Answer: C

Question 8

The function $f(x)$ is defined by:

$$f(x) = \frac{x^2 + ax + b}{2x + c}$$

where a , b , and c are constants.

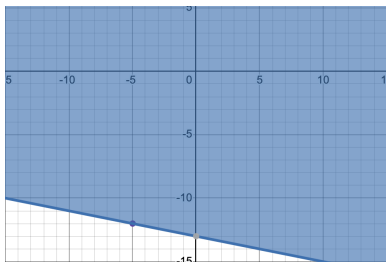
The graph of $f(x)$ does not intersect the line $x = 4$. If $f(5) = f(6) = 0$, what is the value of $a + b + c$?

- A. 10
- B. 11
- C. 12
- D. 13

Answer 8

Correct Answer: B

Question 9



The shaded region represents the solutions to the inequality:

$$rx + ty \geq -65$$

What is the value of $r + t$?

- A. 6
- B. 4
- C. -4
- D. -5

Answer 9

Correct Answer: A

Thank You!



We appreciate your time and focus.

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